

BQI Architecture Diagrams:

Transformer and Neural-Network Circuit Realisation of the Brain Query Interface (BQI) Model

Dung Vu
Phenikaa University

Overview

The figures below translate the formal BQI operators Φ_E (Query Encoder), Σ (SNIS Attention/Search) and Φ_D (Perception Decoder) – together with the hierarchical Bayesian Predictive-Coding (BPC) recurrence and the Dynamic SNIS Network (DSN) – into concrete *circuit-style* neural-network / Transformer block diagrams. Each diagram is drawn with **TikZ** as a signal-flow (“circuit”) schematic: solid arrows carry forward activations (queries, keys, values, predictions), dashed red arrows carry feedback / control / error signals.

1 Top-Level BQI Master Circuit

Figure 1 realises the BQI Master Equation

$$\text{Perception}(t) = \Phi_D \left(\text{softmax} \left(\frac{W_Q \Phi_E(B(t), C(t)) \cdot S^\top}{\sqrt{d_k}} \right) S \right), \quad \frac{d\rho}{dt} = \mathcal{L}_{\text{Lindblad}}(\rho)$$

as a closed-loop circuit. The brain state $B(t)$ and context $C(t)$ enter a Transformer-style **Query Encoder** Φ_E (self-attention + FFN stack), producing the query tensor $Q = W_Q \Phi_E(B, C)$. This query is correlated with the **Supra-Neural Information Space** S (acting as the Key/Value memory bank) through the **SNIS Attention** block Σ , an exact scaled dot-product attention circuit. The resulting answer A is projected by the **Perception Decoder** Φ_D (FFN/decoder stack) to produce $\text{Perception}(t)$. A DMN-gating controller monitors A_{DMN} and Φ and feeds back to widen the projector P_B (rank expansion), i.e. it modulates the effective bandwidth of the Σ block – the circuit-level analogue of meditation / Eq. (15).

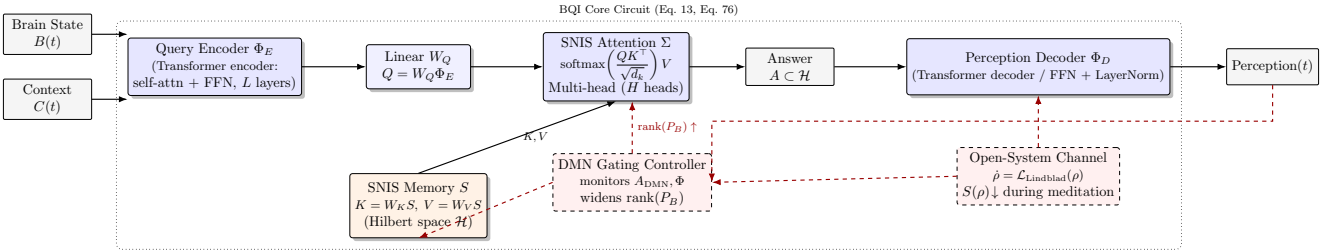


Figure 1: Top-level BQI master circuit. The Query Encoder Φ_E , SNIS Attention Σ and Perception Decoder Φ_D form an encoder–memory–decoder pipeline isomorphic to a Transformer cross-attention block, with the SNIS memory S playing the role of a frozen (or slowly evolving) Key/Value bank. The DMN-gating controller and open-quantum-system channel form feedback loops that modulate the effective bandwidth (rank of P_B) of the Σ block.

2 SNIS Multi-Head Attention Circuit (Detailed)

Figure 2 expands the Σ block of Figure 1 into the exact scaled dot-product / multi-head attention circuit of Eqs. (16)–(20), with each of the H heads corresponding to a distinct cortical processing stream (visual, auditory, semantic, proprioceptive, ...).

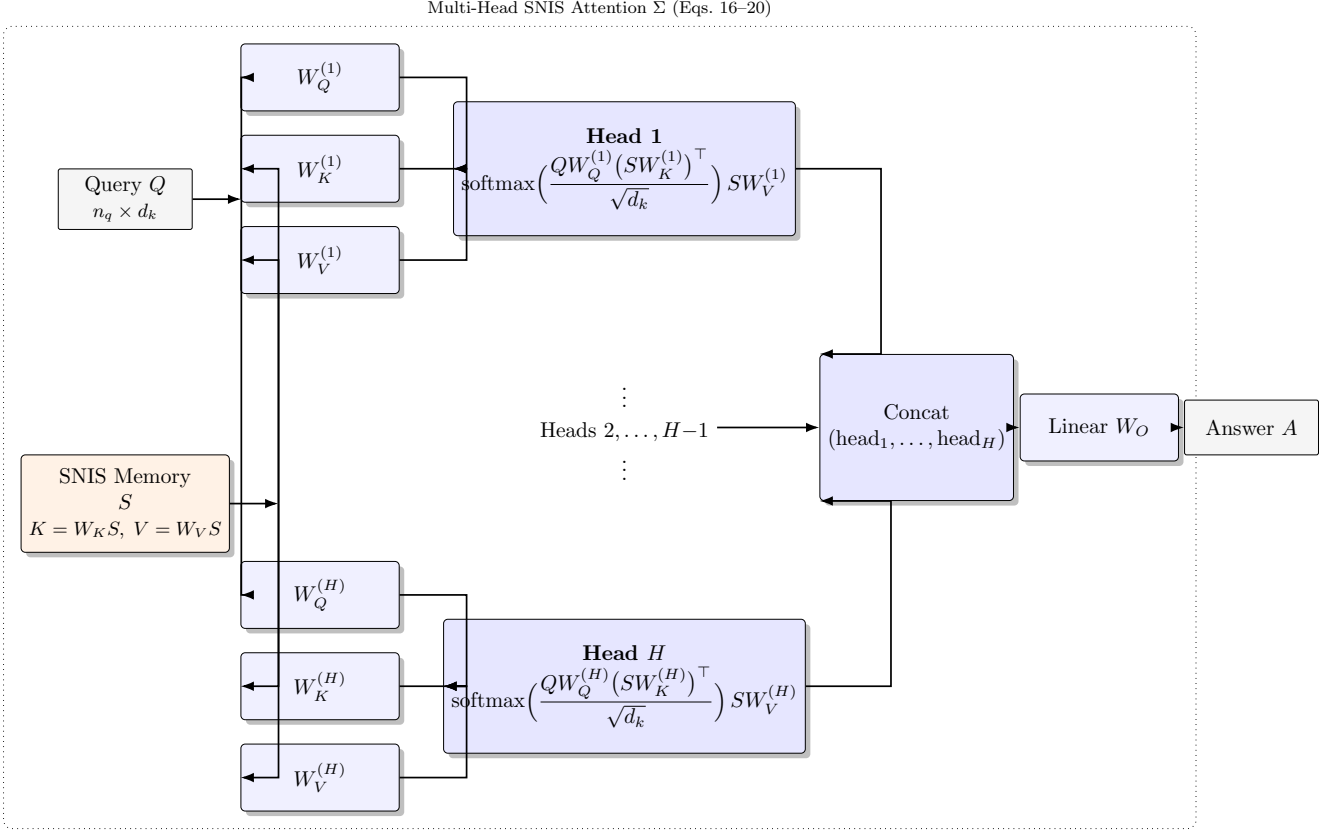


Figure 2: Detailed multi-head SNIS attention circuit. Each head i projects the query Q and the SNIS memory S into a d_k -dimensional subspace via W_i^Q, W_i^K and computes scaled dot-product attention, retrieving a weighted combination of $V = SW_i^V$. The H head outputs are concatenated and linearly mixed by W_O to form the answer A – the biological interpretation assigns each head to a distinct cortical processing stream.

3 Hierarchical BPC Recurrence as a Predictive-Coding RNN

Figure 3 draws the hierarchical Bayesian Predictive-Coding (BPC) recurrence of Eqs. (5)–(7) and (33)–(34) as a stack of L leaky-integrator RNN cells (the Φ_E block of Figure 1 expanded internally). Each level l holds a belief state $\mu^{(l)}$, receives a top-down prediction $f^{(l)}(\mu^{(l+1)})$ from the level above and a bottom-up precision-weighted prediction error $\varepsilon^{(l-1)}$ from the level below; synaptic weights $\theta^{(l)}$ are updated by the local Hebbian/STDP rule of Eq. (34).

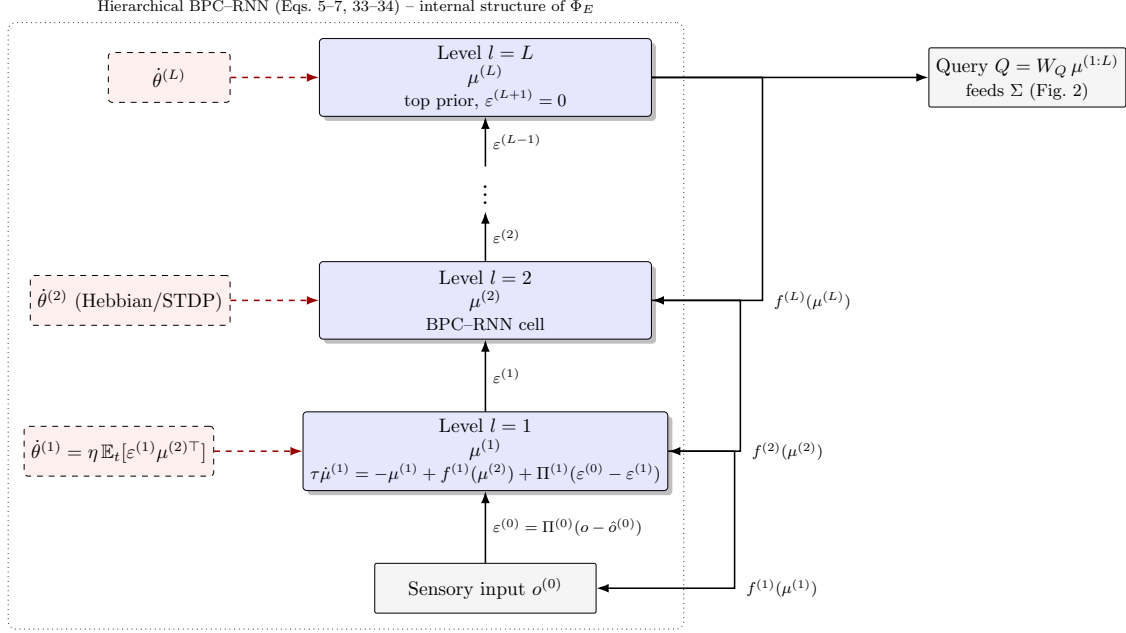


Figure 3: Hierarchical predictive-coding recurrence drawn as a stack of L leaky-integrator RNN cells. Solid upward arrows carry precision-weighted prediction errors $\varepsilon^{(l-1)}$; solid downward arrows (right path) carry top-down generative predictions $f^{(l)}(\mu^{(l+1)})$. Dashed red arrows show local synaptic-plasticity (Hebbian/STDP) updates $\dot{\theta}^{(l)}$ that perform gradient descent on the free energy F . The top-level belief $\mu^{(L)}$ is linearly projected to form the query Q that enters the SNIS attention circuit of Figure 2.

4 Dynamic SNIS Network (DSN): Cognitively-Inspired AI Circuit

Figure 4 realises the DSN architecture of Section 12.3 / Figure 23 of the paper as a complete neural-network circuit: a Transformer backbone acting as Φ_E , a dual short-/long-term memory bank acting as the SNIS proxy, and a hippocampal-consolidation update rule $M_{t+1} = (1 - \delta)M_t + \delta M_{\text{new}}(x_t)$ that lets the memory – unlike frozen LLM weights Θ – evolve with experience.

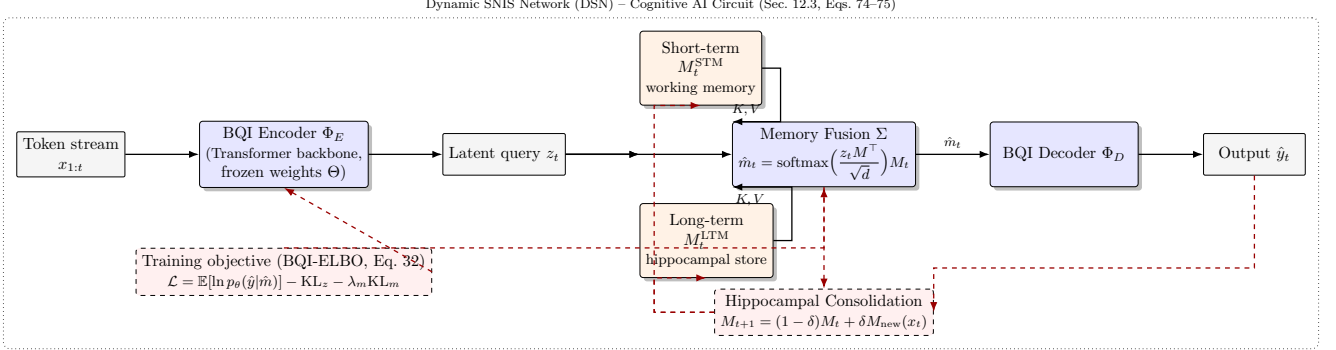


Figure 4: Dynamic SNIS Network (DSN) circuit. A Transformer backbone (BQI Encoder Φ_E) encodes the token stream into a latent query z_t , which attends jointly over a short-term working-memory bank M_t^{STM} and a long-term hippocampal-style memory bank M_t^{LTM} via the Memory Fusion attention block Σ . The fused representation $\hat{m}_t$ is decoded by Φ_D to produce $\hat{y}_t$. Unlike frozen LLM weights Θ , the memory banks are updated online by the hippocampal consolidation rule (dashed red feedback), and the whole circuit is trained end-to-end under the BQI-ELBO objective (Eq. 32).

5 Closed-Loop COM / BCI Circuit

Finally, Figure 5 embeds the BQI core circuit (Figure 1) inside the closed-loop Consciousness Optimisation Model (COM) / BCI pipeline of Algorithm 2 and Figure 21, showing how neural sensing, the BQI inference engine, and stimulation form a feedback control loop driving $A_{DMN} \downarrow$, $\Phi \uparrow$, $QoC \uparrow$.

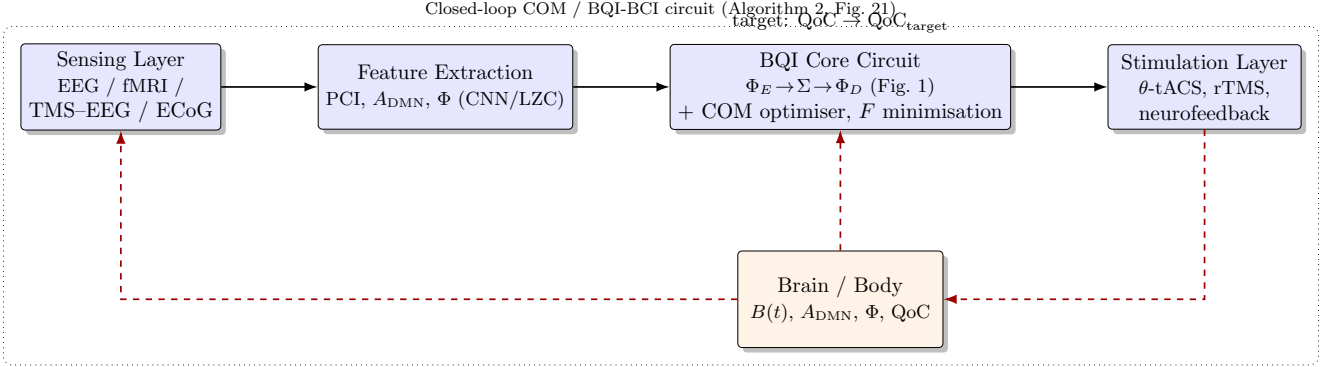


Figure 5: Closed-loop BQI-BCI circuit (Algorithm 2 / COM). The BQI core circuit of Figure 1 sits between a multimodal sensing/feature layer and a stimulation layer (dashed = feedback/control signals), forming a control loop that drives the brain state toward $A_{DMN} \downarrow$, $\Phi \uparrow$ and the target Quality of Consciousness QoC_{target} .